

Assignment 2

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Ex 1

$$\ln y = \ln(C_1 e^{C_2 t}) = \ln C_1 + C_2 t$$

$$\ln y = K + C_2 t$$

$$K + C_2(1) = \ln(2,250)$$

$$K + C_2(2) = \ln(2,500)$$

$$K + C_2(4) = \ln(5,000)$$

$$K + C_2(8) = \ln(29,000)$$

$$K + C_2(12) = \ln(120,000)$$

$$K + C_2(15) = \ln(275,000)$$

$$K + C_2(19) = \ln(1,180,000)$$

$$K + C_2(23) = \ln(3,100,000)$$

$$K + C_2(27) = \ln(7,500,000)$$

$$K + C_2(29) = \ln(24,000,000)$$

$$K + C_2(30) = \ln(42,000,000)$$

$$K + C_2(32) = \ln(220,000,000)$$

$$K + C_2(33) = \ln(410,000,000)$$

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$A =$	1	1	and $b =$	7.710
	1	2		7.824
13×2	1	4		8.517
	1	8		10.275
	1	12		11.695
	1	15		12.523
	1	19		13.981
	1	23		14.946
	1	27		15.830
	1	29		17.0
	1	30		17.55
	1	32		19.21
	1	33		19.83

$$A^T = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 & 12 & 15 & 19 & 23 & 27 & 29 & 30 & 32 & 33 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 13 & 235 \\ 235 & 5927 \end{bmatrix}, \quad A^T B = \begin{bmatrix} 176.90 \\ 3793.23 \end{bmatrix}$$

$$\begin{bmatrix} 13 & 235 \\ 235 & 5927 \end{bmatrix} \begin{bmatrix} k \\ c_2 \end{bmatrix} = \begin{bmatrix} 176.90 \\ 3793.23 \end{bmatrix}$$

$$k = 7.197, \quad c_2 = 0.354, \quad c_7 = c^k = 1335.418$$

$$Y = 1335.418 e^{0.354t}$$

Elsalam

Ex 2

Number of mathematics majors: $M = 18$

Computer science majors: $CS = 325$

$$a) 18 \times 325 = 5850$$

$$b) \text{total} = 18 + 325 = 343$$

$$\text{ways to pick 1st student} = 343$$

Ex 3

$$\text{length} = 8 \text{ to } 12$$

can be contain:

26 lowercase letters

26 uppercase

10 digits

6 special characters

total possible characters per positions

$$c = 26 + 26 + 10 + 6 = 68$$

$$a) \text{total passwords} = 68^8 + 68^9 + 68^{10} + 68^{11} + 68^{12}$$

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b) total possible characters without special

$$\text{characters} = 68 - 6 = 62$$

total passwords with no special characters

$$= 62^8 + 62^9 + 62^{10} + 62^{11} + 62^{12}$$

total with at least one special =

$$= (62^8 + 62^9 + 62^{10} + 62^{11} + 62^{12}) \times$$

$$(62^8 + 62^9 + 62^{10} + 62^{11} + 62^{12})$$

Ex 4

$$n = 100$$

$$\sum_{k=0}^{100} \binom{100}{k}$$

$$k=3$$

$$\binom{100}{0} + \binom{100}{1} + \binom{100}{2} + \dots + \binom{100}{100}$$

$$8251 - 2^{100}$$

$$2^{100} - 8251 = 1.267 \times 10^{30}$$

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Ex 5

$$n=5$$

$$\text{total} = 5! = 120$$

$$\text{one correct order} = 1$$

$$P(\text{in order}) = \frac{1}{120}$$

Ex 6

11 red socks

8 black socks

$$n=19$$

$$k=2$$

$$\text{total} = C(19, 2) = \frac{19!}{2!(19-2)!} = 171 \text{ ways}$$

$$a. C(11, 2) = 55 \text{ ways}$$

$$P(\text{pair of red}) = \frac{55}{171} = 0.322$$

$$b. C(8, 2) = 28 \text{ ways}$$

$$P(\text{pair of black}) = \frac{28}{171} = 0.164$$

$$C(16, 1) \times C(8, 1) = \frac{88}{177} = 0.575$$

$$C(19, 2)$$

I don't know :-

Ex 7

$$\mu = \sum x \cdot P(x)$$

$$\sigma^2 = \sum [x^2 \cdot P(x)] - \mu^2$$

X	P(x)	x ²	x ² · P(x)	x · P(x)
0	0.90	0	0	0
1	0.05	1	0.05	0.05
2	0.03	4	0.12	0.06
3	0.02	9	0.18	0.06

$$\mu = 0 + 0.05 + 0.06 + 0.06 = 0.17$$

$$\sigma^2 = 0 + 0.05 + 0.12 + 0.18 - (0.17)^2 = 0.3277$$

$$\sigma = \sqrt{0.3277} = 0.57$$

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Ex 8

X	P(X)	X · P(X)	X ² · P(X)
4	0,4	1,60	6,40
5	0,3	1,50	7,50
6	0,1	0,60	3,60
8	0,15	1,20	9,60
10	0,05	0,50	5,00

$$\mu = 1,60 + 1,50 + 0,60 + 1,20 + 0,50 = 5,4$$

$$\sigma^2 = 6,4 + 7,5 + 3,6 + 9,6 + 5 - 5,4^2 = 2,94$$

$$\sigma = \sqrt{2,94} = 1,715$$

$$z \cdot P(A) = P(6) + P(8) + P(10) = 0,1 + 0,15 + 0,05 = 0,30$$

$$P(\text{three days}) = (0,30)^3 = 0,027 = \% 2,7$$

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Ex 9

$$P(6) = \frac{2}{3} P(4) \quad \checkmark \quad P(4) + P(6) = 0.575, \quad P(4) = 0.345, \quad P(6) = 0.23$$

x	$P(x)$	$x \cdot P(x)$	$x^2 \cdot P(x)$
1	0.230	0.230	0.230
2	0.180	0.360	0.720
4	0.345	1.380	5.520
6	0.230	1.380	8.280
9	0.015	0.135	1.215

$$\mu = 3.485$$

$$\sigma^2 = 3.819$$

$$\sigma = 1.9544$$

Elsalam

* $Exp = a, b, c$

because they satisfy the Four Conditions

1. fixed number of trials

2. two possible outcomes

3. independent trials

4. constant probability of success

Ex 11

$$n = 8$$

$$p = 0.77, q = 0.23$$

$$P(x) = \frac{n!}{(n-x)!x!} \times p^x \times q^{n-x}$$

$$P(8) = \frac{8!}{(8-8)!8!} \times (0.77)^8 \times (0.23)^{8-8}$$

$$= 0.1235 \approx 12.36\%$$

$$P(x > 4) = P(5) + P(6) + P(7) + P(8)$$



$$* P(5) = {}^8C_5 \times (0.77)^5 \times (0.23)^2 = 0.18443$$

$$P(6) = {}^8C_6 \times (0.77)^6 \times (0.23)^2 = 0.30872$$

$$P(7) = {}^8C_7 \times (0.77)^7 \times (0.23)^1 = 0.29529$$

$$P(8) = 0.12357$$

$$P(X > 4) = 0.91201 = 91.20\%$$

$$\textcircled{C} P(3) = {}^8C_3 \times (0.77)^3 \times (0.23)^5 = 0.1646$$

$$= 16.46\% \approx 1.65\%$$

Ex 12

$$n = 5, P = 0.40, q = 0.60$$

$$\textcircled{a} P(2) = {}^5C_2 \times (0.40)^2 \times (0.60)^3$$

$$= 0.3456$$

$$\textcircled{b} P(X \leq 3) = P(0) + P(1) + P(2) + P(3)$$

$$P(0) = 0.0778, P(1) = 0.2592, P(2) = 0.3456$$

$$P(3) = 0.2304$$

$$P(X \leq 3) = 0.9130$$

$$* \textcircled{c} P(X \geq 2) = P(X) - P(X < 2) = 1 - (P(0) + P(1))$$

$$= 1 - 0.3370 = 0.6630$$

$$\textcircled{d} P(X < 3) = P(0) + P(1) + P(2)$$

$$= 0.0778 + 0.2592 + 0.3456 = 0.6826$$

$\textcircled{13}$

$$\lambda = 4, x = 5$$

$$P(5; 4) = \frac{e^{-4} 4^5}{5!} = 0.15629$$

$$\textcircled{14} P(5) = 0.32 (4 - 0.32)^{5-1} = 0.6842$$

$\textcircled{15}$

$$\textcircled{a} A_{left} = 1 - 0.4887 = 0.113, z = -2.28$$

$$\textcircled{b} A_{left} = 1 - 0.8213 = 0.1788, z = -0.92$$

$$\textcircled{c} 1 - 0.6064 = 0.3936, z = -0.27$$

$\textcircled{16}$

$$\textcircled{a} \frac{\sigma}{2} = \frac{0.05}{2} = 0.025, -z = -1.96, +z = +1.96$$

$$\textcircled{b} \frac{\sigma}{2} = 0.05, -z = -1.645, +z = 1.645$$

$$\textcircled{c} \frac{\sigma}{2} = 0.005, -z = -2.58, +z = 2.58$$

Ex 17

$$\mu = 225, \sigma = 10$$

$$a) P(200 < x < 220) = P(200 - 225 < z < 220 - 225)$$

$$\left(\frac{200 - 225}{10} < z < \frac{220 - 225}{10} \right)$$

$$P(-2.5 < z < -0.5) = 0.3085 - 0.0062$$

$$= 0.3023$$

$$b) P(x < 200) = P(z < \frac{200 - 225}{10}) = 0.0062$$

Ex 18

$$\mu = 949, \sigma = 100$$

$$a) P(x > 1200) = P(z > \frac{1200 - 949}{100})$$

$$= P(z > 2.51)$$

$$P(z < 2.51) = 0.9940$$

$$P(x > 1200) = 1 - P(z < 2.51) = 1 - 0.9940$$

$$= 0.0060$$

* ② $Z = -1.28$ from the table

$$X = Z \cdot \sigma + \mu = (-1.28 \times 100) + 999 = 821$$

Ex 19

$$\mu = 4.8, \sigma^2 = 2.1, \sigma = 1.4497$$

①

$$90\% = \text{area } 0.8997, Z = 1.28 \text{ (the closest)}$$

$$X = (1.28 \times 1.4497) + 4.8 = 6.65 \text{ days}$$

② $Z = -0.52$

$$X = 4.05$$

Ex 20

$$\mu = 215, \sigma = 15$$

$$Z = \frac{220 - 215}{15} = 0.33 \text{ (a)}$$

$$P(Z < 0.33) = 0.6293$$

$$P(X > 220) = 1 - 0.6293 = 0.3707$$



$$\textcircled{b} \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}, n = 25$$

$$\sigma_{\bar{x}} = 3, z = \frac{220 - 215}{3} = 1.67$$

$$P(z < 1.67) = 0.9525$$

$$P(\bar{x} < 220) = 1 - 0.9525 = 0.0475$$

Ex 21

$$n = 32, \sigma = 165.1, \alpha = 1 - 0.92 = 0.08$$

$$\bar{x} = \frac{11,080}{32} = 346.25$$

$$\text{area} = 1 - 0.04 = 0.96$$

$$z = 1.75$$

$$E = \frac{z_{\alpha}}{2} \times \left(\frac{\sigma}{\sqrt{n}} \right) = 1.75 \times \left(\frac{165.1}{\sqrt{32}} \right) = 51.07$$

$$\bar{x} - E < \mu < \bar{x} + E = 346.25 - 51.07 < \mu < 346.25 + 51.07$$

$$295.18 < \mu < 397.32$$

* Ex 22

① $H_0: \mu = 24.6, H_1: \mu \neq 24.6$

② $H_0: \mu = 57,497, H_1: \mu \neq 57,497$

③ $H_0: \mu = 25.4, H_1: \mu > 25.4$

④ $H_0: \mu = 88, H_1: \mu < 88$

⑤ $H_0: \mu = 70, H_1: \mu < 70$

